Shubhang Kulkarni

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.74
Mahesh PU College	Class XII (Karnataka School Examination and Assessment Board (KSEAB))	2024	96.30%
Sri Vidya Kendra, The Smart School	Class X (Central Board of Secondary Education)	2022	97.00%

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award, Indian Institute of Technology Hyderabad

- Secured the **highest GPA (9.75)** in the Electrical Engineering (ICDT) branch for calendar year 2025.

SKILLS

- **Hardware Description Languages (HDLs)**: Verilog, Assembly (ARM, RISC-V ISA)
- **Programming Languages:** C, Python, MATLAB
- **Circuit Design and Simulation Tools**: Altium Designer, KiCad, LTspice, Icarus Verilog, GTKWave

RELEVANT COURSES

Circuits and Networks Analysis, Digital Systems, Engineering Electromagnetics, Semiconductor Device Fundamentals, Numerical Methods for Device Modelling, Analog Electronics/Lab, PCB Design and Layout for Electronic Systems, Microprocessor and Computer Architecture

PROJECTS

LCR Meter (In progress) [Summary] | Altium Designer, *Analog Front-End, PCB Layout*

- Engineered the complete **Analog Signal Chain** for MCU-based impedance measurement, implementing AC-coupling, active low-pass filtering to **suppress DAC stair-step harmonics**, and DC level-shifting to optimise ADC input range.
- Implemented impedance measurement using sinusoidal excitation and **Quadrature Phase Detection (IQ Method)** across selectable range resistors.
- Collaborated on the PCB design and layout, ensuring the analog signal paths from the Device Under Test (DUT) and the sensing circuitry were routed carefully to avoid noise.

Compact Air Monitor (In progress) [GitHub] | *Altium, Power Hardware, PCB Design*

- Designed a **compact (6cm x 4cm x 2cm)** two-layered PCB around the ESP32-C3, integrating I2C environmental sensor arrays and user input-output peripherals.
- Handled the power supply by incorporating a **3.3V buck converter** module, with **active bypass battery charging**.
- Executed high-density component placement and PCB trace routing, adhering to **Design for Manufacturability** and the DRC rules of the PCB vendor.
- Collaborated on the mechanical enclosing of the device, defining form-factor constraints and sensor opening requirements for ideal measurement.

RV32I 5-Stage Pipelined Core [GitHub] | *Verilog, Icarus Verilog, GTKWave, RTL Design*

- Designed a **5-stage pipelined RV32I** processor core (IF-ID-EX-MEM-WB) in Verilog, with hazard handling via MEM/WB-to-EX **forwarding** and internal register file bypassing.
- Implemented a **Hazard Detection Unit** to execute pipeline stalls and flushing, resolving load-use data hazards and control hazards caused by branch misprediction and jumps.
- Integrated **exception handling** using CSRs (mtvec, mepc, and mcause) to handle software interrupts and instruction traps across stages.
- Implemented a **GShare Branch Predictor** (10-bit Global History Register, 1024-entry PHT) with a 128-entry direct-mapped Branch Target Buffer (BTB).
- Verified functionality against the **official riscv-tests suite** and validated timing waveforms on GTKWave.

1D FEM Schrödinger Quantum Solver [GitHub] | *C, Numerical Methods, Device Physics*

- Implemented the 1D Schrödinger solver in C, using **Finite Element Method (FEM)** to model quantum confinement.
- Implemented **Inverse Iteration** and **Gauss-Seidel** matrix solvers to compute energy eigenvalues and wavefunctions for various potential barrier profiles.
- Formulated FEM matrices and enforced boundary conditions, validating against analytical quantum well solutions.

POSITIONS OF RESPONSIBILITY

UG Mentor - Sunshine IITH

- Conducted mentor-mentee meets, as a mentor, for the incoming batch of the students.
- Discussed and guided the mentees through challenges, academic and otherwise.

Senior Core - Elektronica

- Collaborating within a team to implement custom hardware accelerators.
- Built the foundational RV32I core from scratch to serve as a primary processing unit for integrating and testing future accelerator modules.

EXTRACURRICULAR

UG Teaching Assistant - EE1080: Probability

- Conducted tutorial session to assist in solving the practice problem sets.
- Evaluated submissions for the programming assignment problems.